

האוניברסיטה העברית בירושלים
THE HEBREW UNIVERSITY OF JERUSALEM

ON VISUAL NEURAL NETWORKS: ARTIFICIAL & NATURAL

Daniella Har Shalom & Yair Weiss

Gatsby Meeting
June 2024



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BACKGROUND

Gatsby Tricentre 2023

- *“When is Unsupervised Disentangelment Possible?” (Neurips)*

“What do Visual Neural Networks Learn?”

- Annual Review of Vision Science, 2025

WHAT DO VISUAL NEURAL NETWORKS LEARN

- Current models
  - CNNs, Vision Transformers…
  - CIFAR, ImageNet, LAION…
  - SotA accuracy
- Behaviorally – performance/“result”
  - (Internally – representations, “hypothesis”…)
- Compared to humans
  - (Compared to ground truth - accuracy, representation…)

	ImageNet Val.
ResNet50	81.5
ResNet101	82.3
ViTLarge	85.8
ViTBasePretrained21k	84.3
Swin	83.6
BiT (JFT300M)	80.3
DINOv2 (LVD-142M)	84.5
Flava (PMD 70M)	75.5
CLIPViTB32 (400M)	62.9
CLIPViTB32 (2B)	66.6
CLIPViTL14 (400M)	72.8

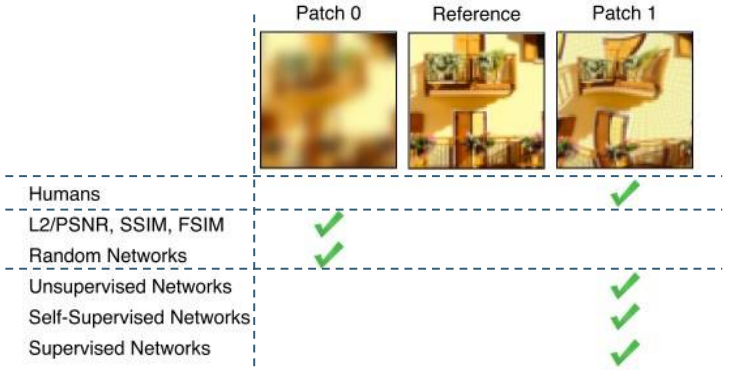
	Acc
ViT-B-16	80.3%
ViT-L-16 (224)	82.0%
ViT-L-16 (512)	85.4%
BiT-M-R50x1	79.9%
BiT-M-R152x4	85.3%
ResNet-50	74.5%
ResNet-152	77.0%

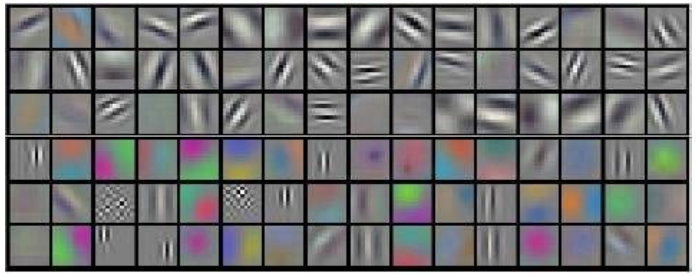
Model	Accuracy
vanila vit	76.25*
Resnet50	76.74
ConvNext-T	78.29
ConvNext-L	**83.47**
open-CLIP	69.33*
DINOv2	**84.11**

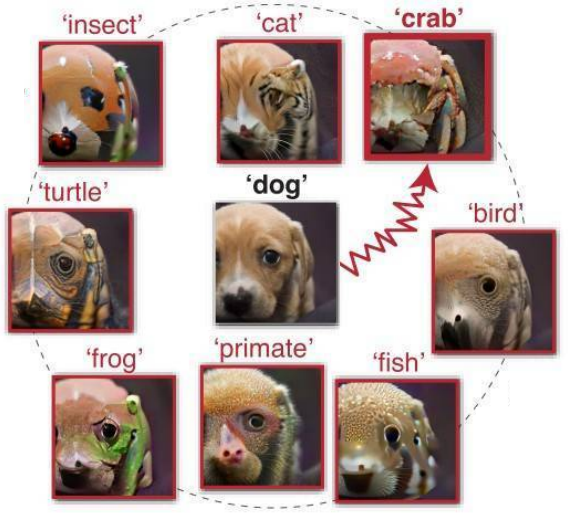
WHAT DO VISUAL NEURAL NETWORKS LEARN

The ***good***

- Remarkable results on different tasks…
- Similarities to the human visual system
  - Gabor filters
  - Perceptual similarity
  - Behavioral correspondence
- Suggested as possible model for human vision







Zhang, R., Isola, P., Efros, A. A., Shechtman, E., & Wang, O. (2018).
Gaziv, G., Lee, M., & DiCarlo, J. J. (2024).

WHAT DO VISUAL NEURAL NETWORKS LEARN

The ***good***

- Remarkable results on different tasks…
- Similarities to the human visual system
- Suggested as possible model for human vision

The ***bad***

- Fooled by changes to which humans are invariant
- Invariant to large changes that disrupt human perception

The ***ugly*** (conclusion?)

- “Good performance” does not necessarily mean “good model”

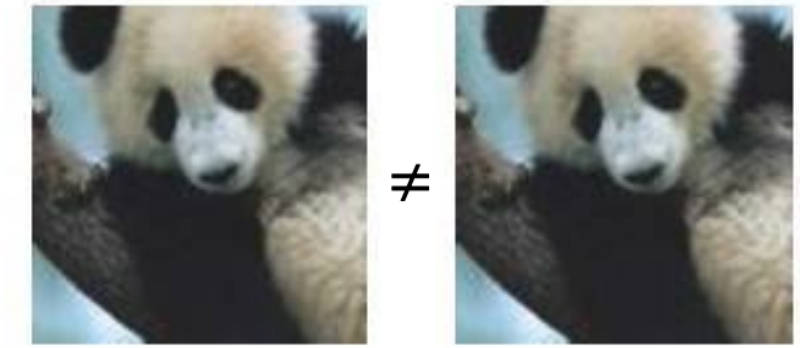
SENSITIVITY AND INVARIANCE

Sensitivity

- To small changes to which humans are invariant
- *“False negatives”*

Invariance

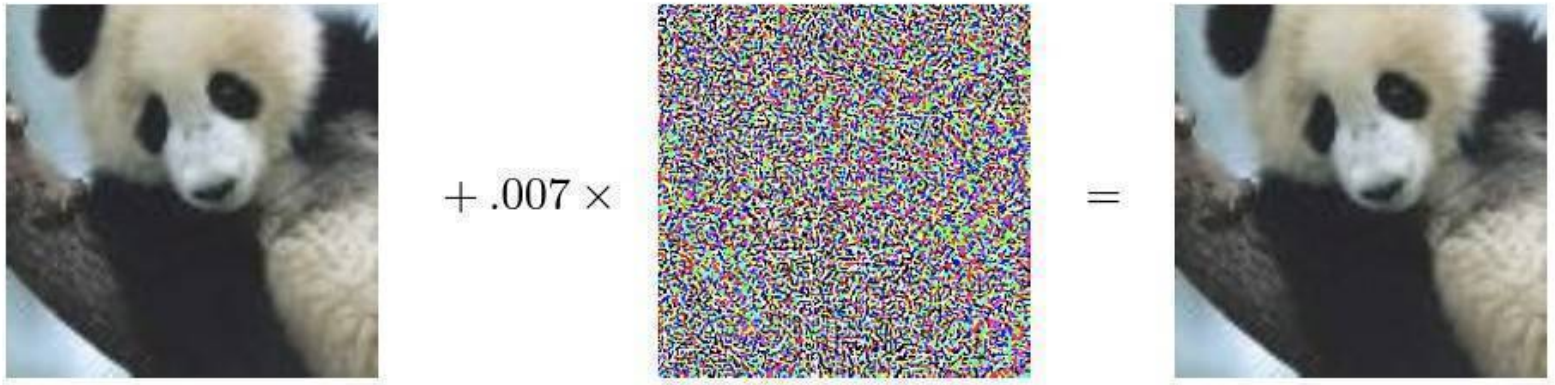
- To large changes that disrupt human perception
- *“False positives”*



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PERTURBATION-BASED CHANGES



“panda”

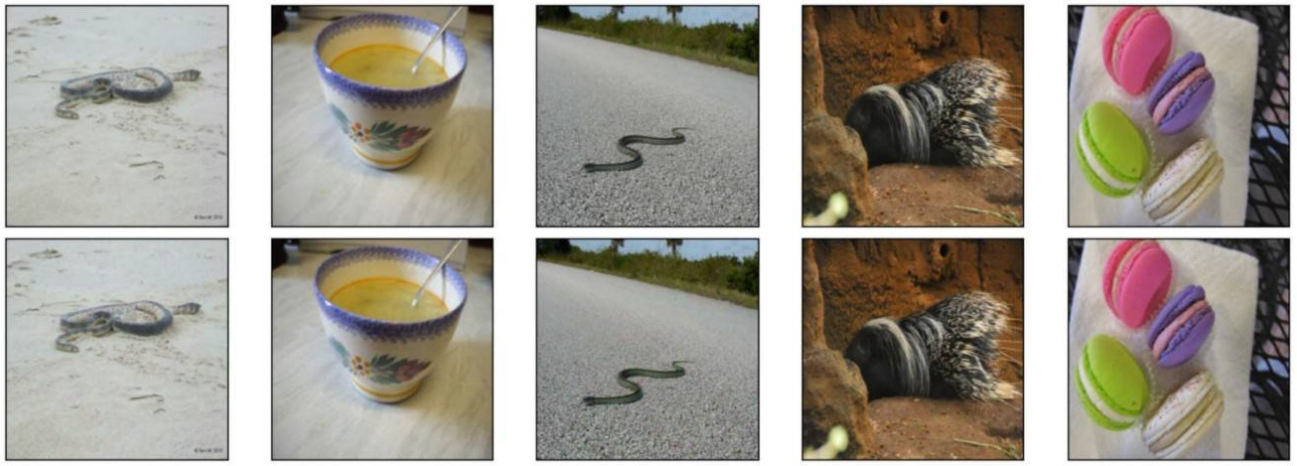
“gibbon”

Goodfellow, I. J., Shlens, J., & Szegedy, C. (2014).

PERTURBATION-BASED CHANGES

Limit change $\left|\left|x - x_{adv}\right|\right|_{\infty} \leq \epsilon$ at each pixel, where $\epsilon = 0.062$.

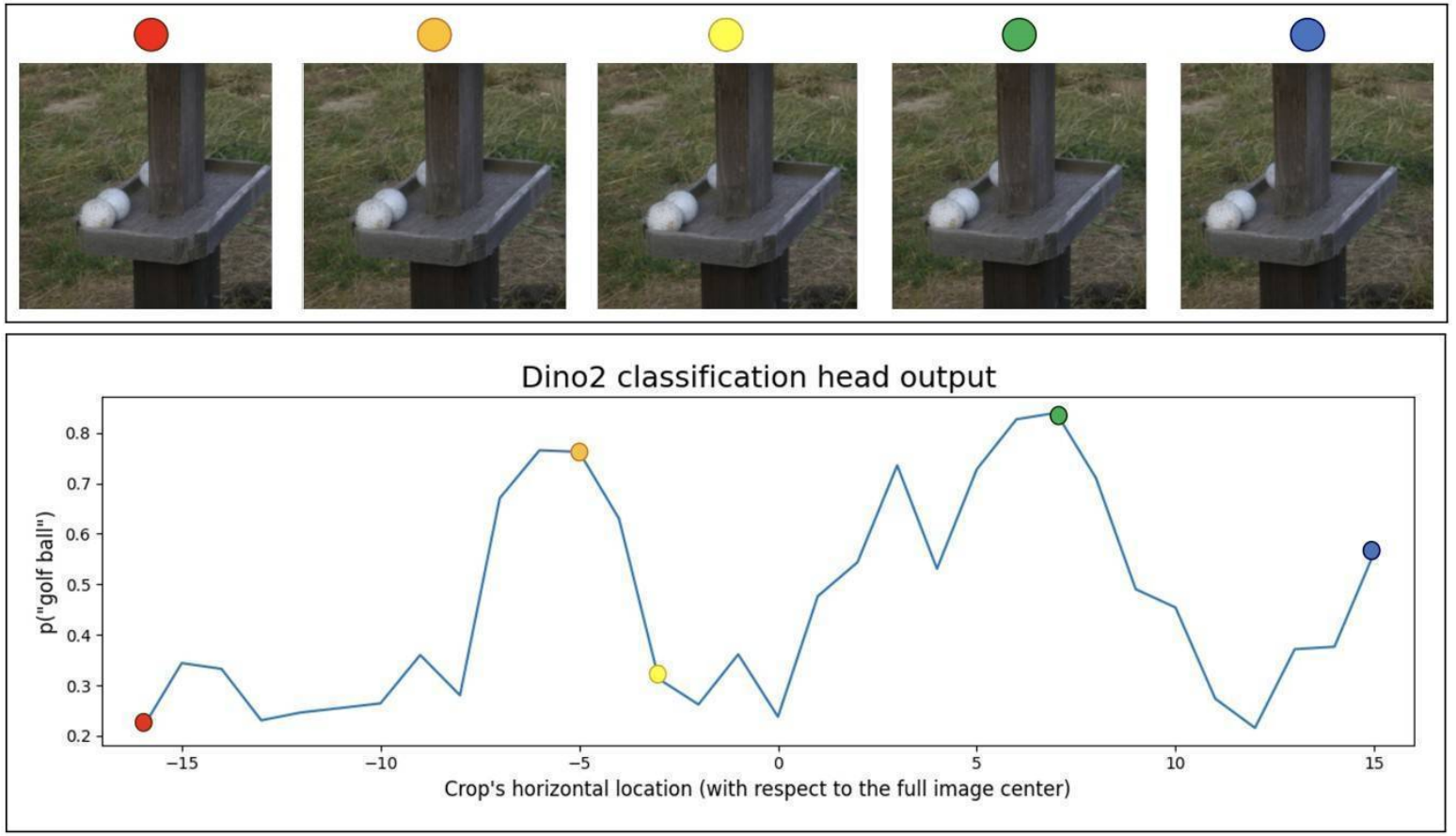
	ImageNet	
	APGD	Acc
ViT-B-16		80.3%
ViT-L-16 (224)		82.0%
ViT-L-16 (512)		85.4%
BiT-M-R50x1		79.9%
BiT-M-R152x4		85.3%
ResNet-50		74.5%
ResNet-152		77.0%



These models can be fooled for 100% of images!

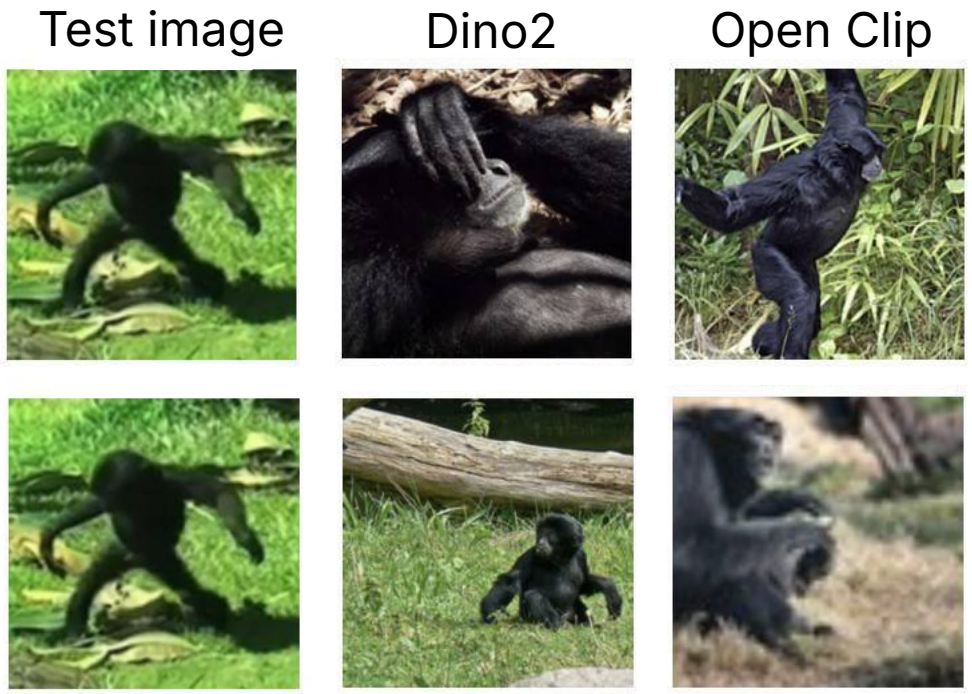
Mahmood, K., Mahmood, R., & Van Dijk, M. (2021).

SMALL IMAGE SHIFT



Shifman, O., & Weiss, Y. (2024).

SMALL IMAGE SHIFT



Model	Accuracy	Adv-Rob (1-NN) Shift Size:
		1
vanila vit	76.25*	37.21
Resnet50	76.74	46.33
ConvNext-T	78.29	59.16
ConvNext-L	**83.47**	**65.58**
open-CLIP	69.33*	57.01
DINOv2	**84.11**	**62.42**

For ~40% of images, these models can be fooled by a 1-pixel shift in the image.

TEXTURE BIAS

	Content image	Texture image	Texture-shape cue conflict
			
ResNet-50			
Humans	Cat	Elephant	Cat

Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2018).

TEXTURE BIAS



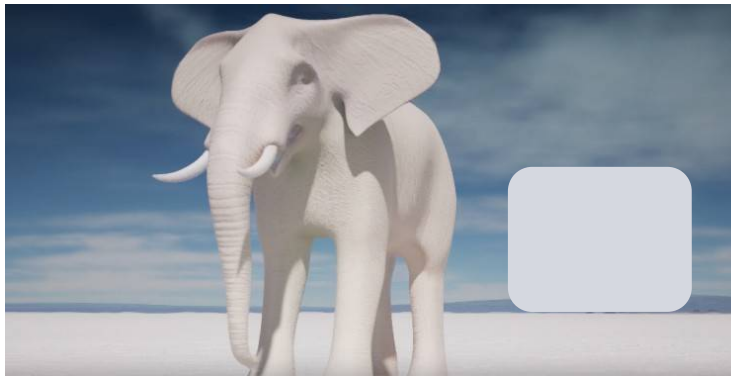
Human observers
AlexNet
VGG- 16
GoogLeNet
ResNet-50

TEXTURE BIAS









Bordes, F., Shekhar, S., Ibrahim, M., Bouchacourt, D., Vincent, P., & Morcos, A. (2024).

PUG: ANIMALS



A front-left view of a medium-sized elephant in a salt flats environment.



A front-left view of a medium-sized elephant textured with sky design, in a salt flats environment.



A front-left view of a medium-sized elephant textured with a grass design in a slat flats environment.



A front-left view of a medium-sized elephant textured with an asphalt design in a salt flats environment.



A front-left view of a medium-sized elephant in an Australian environment.



A front-left view of a small elephant in a futuristic gallery environment.



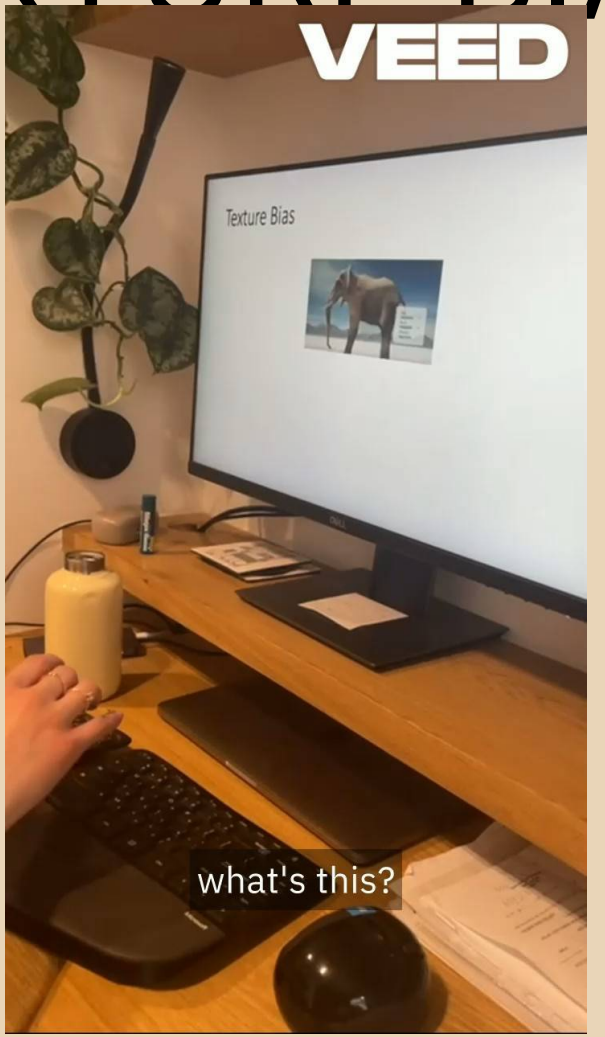
A front-right view of a medium-sized elephant in an opera house environment.



A front-right view of a medium-sized elephant in an outdoor bridge environment.

THEOREM: HUMANS DO NOT EXHIBIT TEXTURE BIAS

Proof Sketch:



■

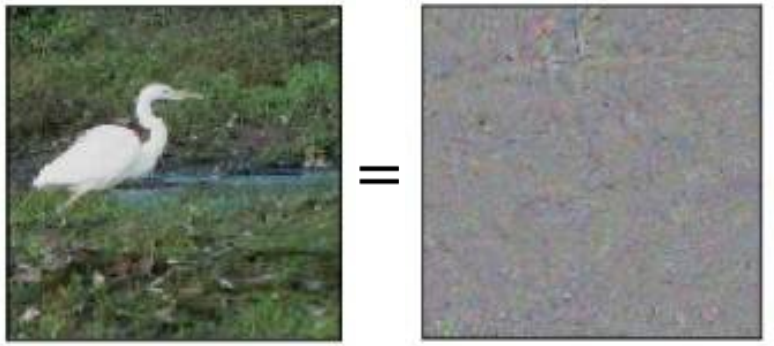
SENSITIVITY AND INVARIANCE

Sensitivity

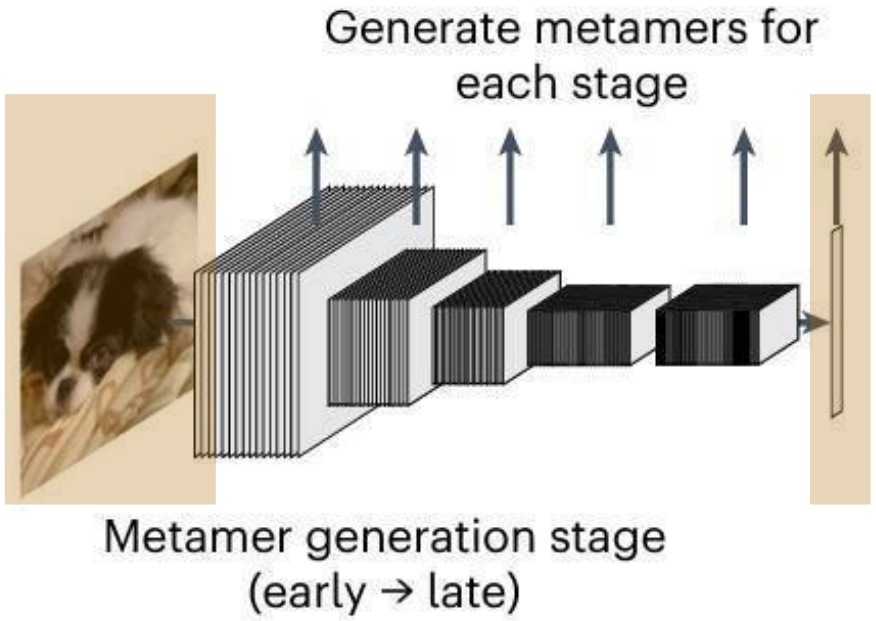
- To small changes to which humans are invariant
- *“False negatives”*

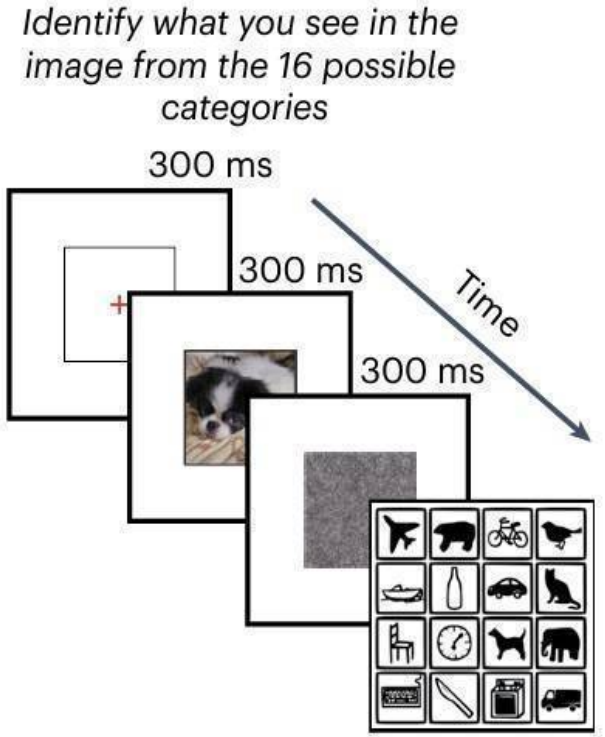
Invariance

- To large changes that disrupt human perception
- *“False positives”*



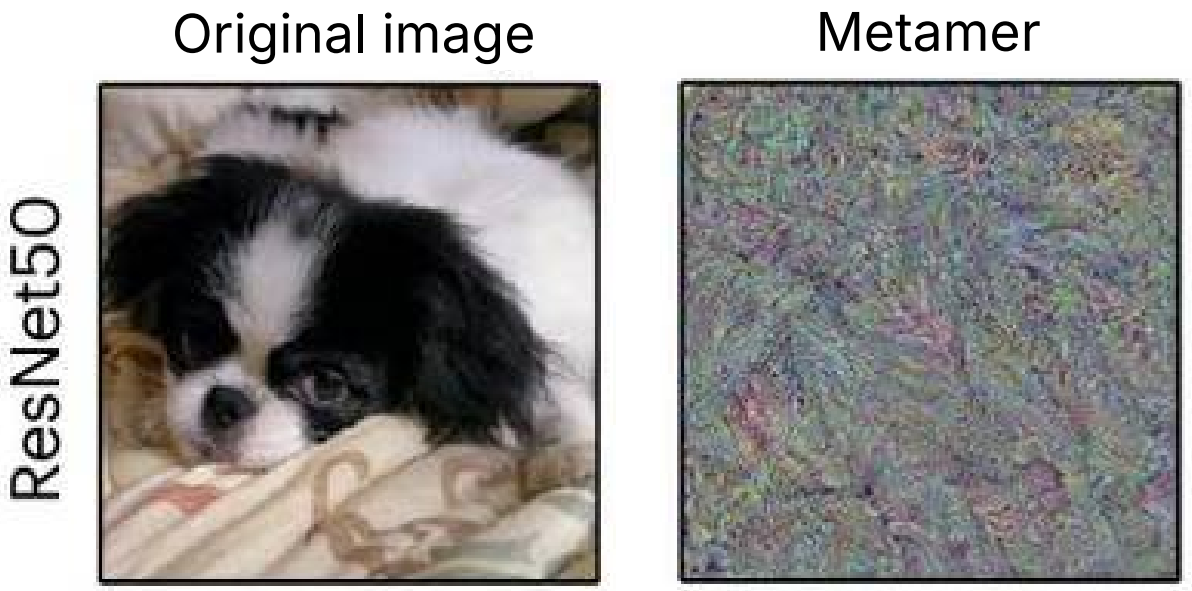
MODEL METAMERS



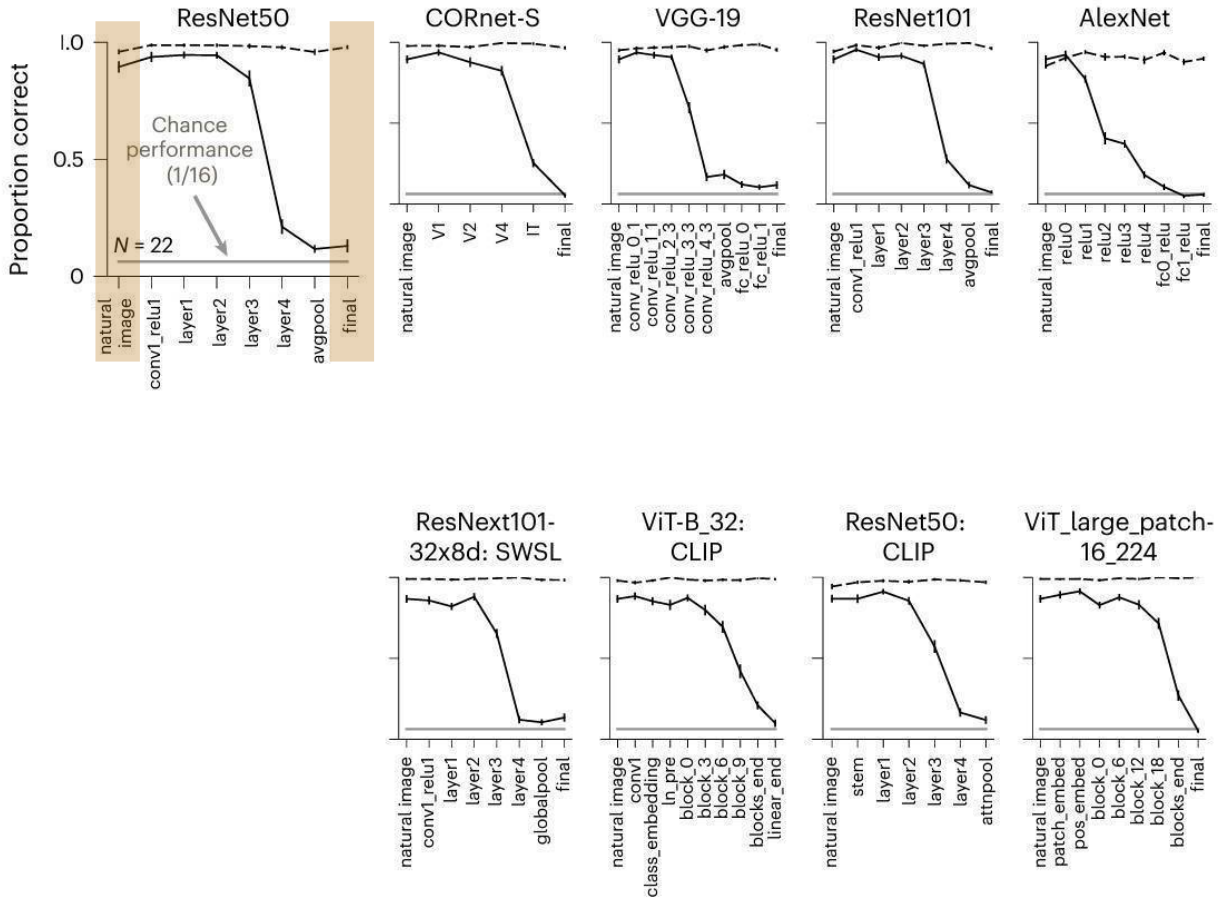


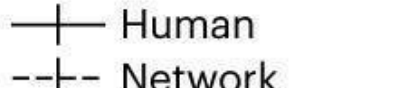
Feather, J., Leclerc, G., Mądry, A., & McDermott, J. H. (2023)

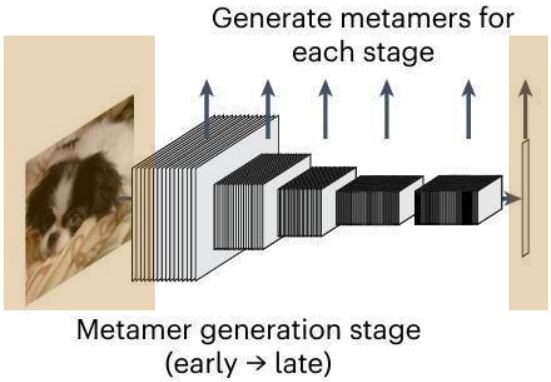
MODEL METAMERS



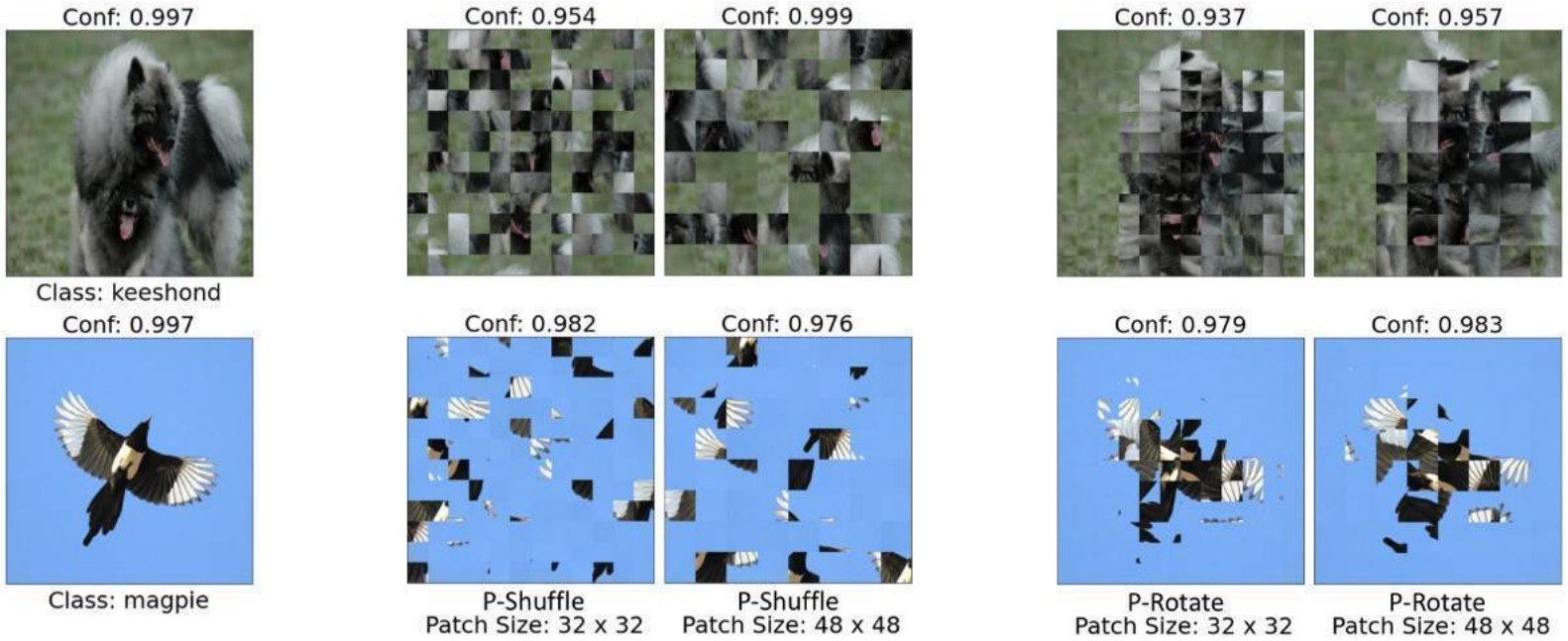
MODEL METAMERS





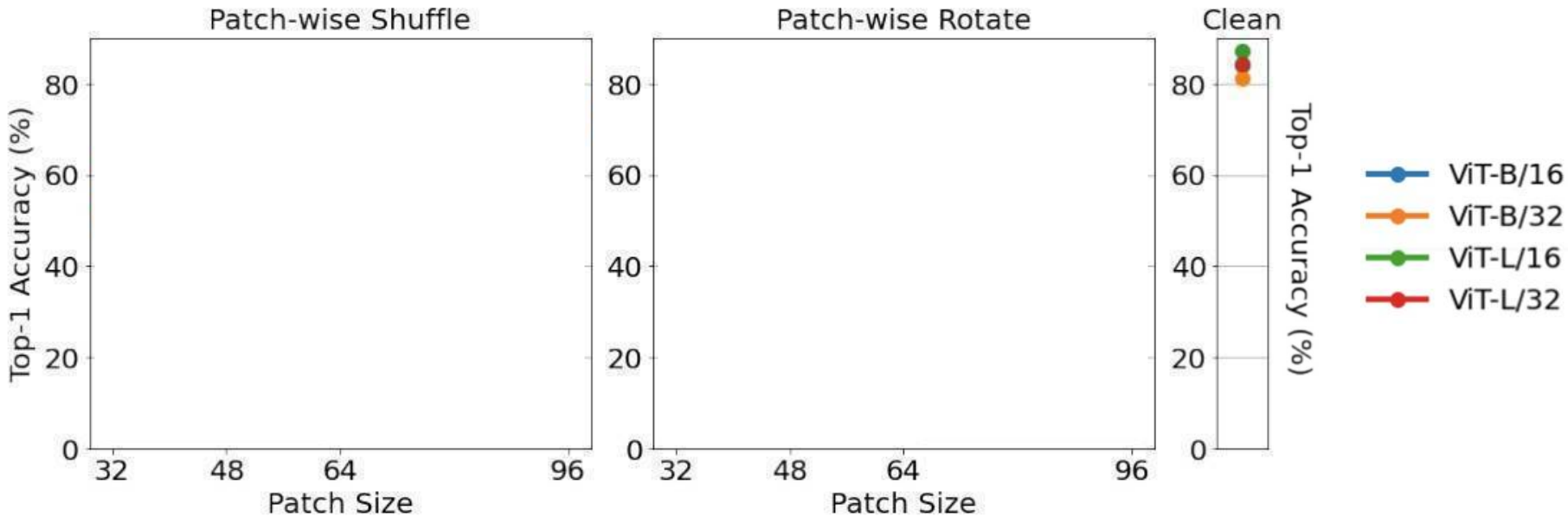


PATCH-BASED TRANSFORMATIONS



Qin, Y., Zhang, C., Chen, T., Lakshminarayanan, B., Beutel, A., & Wang, X. (2022).

PATCH-BASED TRANSFORMATIONS



CONCLUSION

Current models are “good”, and getting better by the minute…

- Suggested as possible model for human vision

Exhibit different invariances

- Sensitivity: fooled by small changes to which humans are invariant
- Invariance: invariant to large changes that disrupt human perception

“Good performance” does not necessarily mean “good model”